

# FINAL SHANK3 ASO Mechanism-of-Action Report

R / targets chemistry-aware bioinformatics workflow

2026-09-15

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# 1 Executive summary

This report summarizes a reproducible **R-only** workflow for two SHANK3-associated ASOs (ASO\_4\_5\_2\_4, ASO\_4\_5\_2\_6) in *Homo sapiens* **GRCh38**. Both ASOs are linked to **increased SHANK3 expression**. Chemistry is **phosphorothioate + 2'-MOE with all cytosines as 5-methyl-dC**; **architecture is unknown** and **RNase-H compatibility is unknown**. Sequence/database evidence is used only to generate **ranked hypotheses**, not confirmed mechanisms.

| metric                    | value                                                |
|---------------------------|------------------------------------------------------|
| Exact SHANK3 mapping rows | 13                                                   |
| Approximate mapping rows  | 1                                                    |
| Top mechanism labels      | ASO_4_5_2_4=RBP occlusion; ASO_4_5_2_6=RBP occlusion |

# 2 ASO sequences and chemistry

| aso_id      | cleaned_sequence   | length | target_percent | chemistry        | backbone | is_methylated | five_methyl_dC | chemistry | rnase_h_compat | was_confirmed |
|-------------|--------------------|--------|----------------|------------------|----------|---------------|----------------|-----------|----------------|---------------|
| ASO_4_5_2_4 | TCCTGACTGTCGCGCGCG | 16     | 100            | phosphorothioate | TRUE     | TRUE          | unknown        | unknown   | NA             | ok            |
| ASO_4_5_2_6 | TAGGCACTAGGCACTAGG | 16     | 100            | phosphorothioate | TRUE     | TRUE          | unknown        | unknown   | NA             | ok            |

**Chemistry rules applied throughout:** do not assume gapmer architecture; do not assume RNase-H activity; annotate every C as 5m-dC while retaining canonical bases for alignment/motifs.

# 3 Target gene and phenotype context

- Gene: **SHANK3** (neuronal synaptic scaffolding; dosage-sensitive ASD/Phelan-McDermid relevant gene).
- Cell types of interest: **neuron, neural precursor cell**.
- Phenotype for both ASOs: **increased SHANK3 expression**.
- Implication: mechanisms that **destroy** SHANK3 mRNA via RNase-H are directionally inconsistent with the stated phenotype for simple on-target knockdown (unless isoform switching/feedback is demonstrated).

# 4 Transcript and genomic mapping (GRCh38)

| aso_id      | match_exact | approx | feature    | transcript      | start | end | strand | mismatches | support_note                                                | status | warning |
|-------------|-------------|--------|------------|-----------------|-------|-----|--------|------------|-------------------------------------------------------------|--------|---------|
| ASO_4_5_2_4 | 1           | 1      | transcript | ENST00000243795 | 16    | 31  | +      | 0          | Exact reverse-complement match in SHANK3 spliced transcript | ok     | NA      |

| aso_id       | match_exact | approx2 | feature_type | transcript         | start | end | strand | mismatches | support_note                                                | status | warning          |
|--------------|-------------|---------|--------------|--------------------|-------|-----|--------|------------|-------------------------------------------------------------|--------|------------------|
| ASO_4x5c2_41 | TRUE        | FALSE   | transcript   | ENST00000167296.3  | 12    | 12  | 0      | 2          | Exact reverse-complement match in SHANK3 spliced transcript | ok     | NA               |
| ASO_4x5c2_42 | TRUE        | FALSE   | transcript   | ENST00000167548.2  | 12    | 12  | 0      | 3          | Exact reverse-complement match in SHANK3 spliced transcript | ok     | NA               |
| ASO_4x5c2_43 | TRUE        | FALSE   | transcript   | ENST00000145200.4  | 14    | 14  | 0      | 4          | Exact reverse-complement match in SHANK3 spliced transcript | ok     | NA               |
| ASO_4x5c2_44 | TRUE        | FALSE   | transcript   | ENST00000163407.2  | 18    | 18  | 0      | 5          | Exact reverse-complement match in SHANK3 spliced transcript | ok     | NA               |
| ASO_4x5c2_45 | TRUE        | FALSE   | transcript   | ENST00000140559.2  | 19    | 19  | 0      | 6          | Exact reverse-complement match in SHANK3 spliced transcript | ok     | NA               |
| ASO_approx41 | FALSE       | TRUE    | transcript   | ENST00000165933.5+ | 15    | 15  | 2      | 7          | Approximate RC match (2 mismatch) in SHANK3 transcript      | ok     | approximate_only |
| ASO_4x5c2_46 | TRUE        | FALSE   | transcript   | ENST00000145785.2  | 20    | 20  | 0      | 1          | Exact reverse-complement match in SHANK3 spliced transcript | ok     | NA               |

| aso_id        | match_exact | approx2 | feature_type | transcript         | start    | end      | strand | mismatches | support                                                     | note | status | warning |
|---------------|-------------|---------|--------------|--------------------|----------|----------|--------|------------|-------------------------------------------------------------|------|--------|---------|
| ASO_4x5c2_671 | TRUE        | FALSE   | transcript   | ENST00000262754.6  | 16275476 | 16275476 | 0      | 2          | Exact reverse-complement match in SHANK3 spliced transcript |      | ok     | NA      |
| ASO_4x5c2_672 | TRUE        | FALSE   | transcript   | ENST00000262754.6  | 16275476 | 16275476 | 0      | 3          | Exact reverse-complement match in SHANK3 spliced transcript |      | ok     | NA      |
| ASO_4x5c2_673 | TRUE        | FALSE   | transcript   | ENST00000245240.4  | 16275476 | 16275476 | 0      | 4          | Exact reverse-complement match in SHANK3 spliced transcript |      | ok     | NA      |
| ASO_4x5c2_674 | TRUE        | FALSE   | transcript   | ENST00000263929.2  | 16275476 | 16275476 | 0      | 5          | Exact reverse-complement match in SHANK3 spliced transcript |      | ok     | NA      |
| ASO_4x5c2_675 | TRUE        | FALSE   | transcript   | ENST00000265380.0+ | 16275476 | 16275476 | 0      | 6          | Exact reverse-complement match in SHANK3 spliced transcript |      | ok     | NA      |
| ASO_4x5c2_676 | TRUE        | FALSE   | transcript   | ENST00000240593.6  | 16275476 | 16275476 | 0      | 7          | Exact reverse-complement match in SHANK3 spliced transcript |      | ok     | NA      |

Windows ( $\pm 200$  /  $\pm 200$  nt) are written to results/02\_target\_windows.fasta and .bed.

## 5 RNA structure and accessibility

ViennaRNA (RNAfold/RNAPfold) is invoked only via system2(). Missing tools produce warnings and empty numeric fields - **results are never fabricated**. All energies are canonical-RNA approximations unsuitable as exact MOE/PS/5m-dC biochemical measurements.

| aso_id       | window_id | context         | rf_size | structure | clipped | prob | ability | energy                                                      | caveat                                                                                            |
|--------------|-----------|-----------------|---------|-----------|---------|------|---------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| ASO_4_5_2    | self      | ASO_4_5_2       | self    | NA        | NA      | NA   | ok      | NA                                                          | Values are canonical-RNA approximations; not exact biochemical measurements for MOE/PS/5m-dC ASOs |
| ASO_4_5_2    | self      | ASO_4_5_2       | self    | NA        | NA      | NA   | ok      | NA                                                          | Values are canonical-RNA approximations; not exact biochemical measurements for MOE/PS/5m-dC ASOs |
| ASO_4_5_2445 | 2_4       | ENST00000415716 | 196.7   | 0.5799568 | -38.4   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2445 | 2_4       | ENST00000675271 | 200.0   | 0.5799568 | -38.4   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2445 | 2_4       | ENST00000262195 | 200.0   | 0.5799568 | -38.4   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2445 | 2_4       | ENST00000415716 | 196.7   | 0.5799568 | -38.4   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2445 | 2_4       | ENST00000675271 | 200.0   | 0.5799568 | -38.4   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2445 | 2_4       | ENST00000668192 | 200.0   | 0.5799568 | -38.4   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2445 | 2_4       | ENST00000710251 | 200.0   | 0.5799568 | -38.4   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2465 | 2_6       | ENST00000415716 | 196.7   | 0.5799568 | -30.8   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2465 | 2_6       | ENST00000675271 | 199.1   | 0.5799568 | -30.8   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2465 | 2_6       | ENST00000262195 | 199.1   | 0.5799568 | -30.8   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |
| ASO_4_5_2465 | 2_6       | ENST00000415716 | 196.7   | 0.5799568 | -30.8   | ok   | NA      | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |                                                                                                   |

| aso_id | window_id   | context           | size  | structure | chapped | hybrid | energy | energy_caveat                                               |
|--------|-------------|-------------------|-------|-----------|---------|--------|--------|-------------------------------------------------------------|
| ASO_4  | ASO2465_2_6 | ENST00000664402.2 | 102.2 | 102.2     | 102.2   | 102.2  | 102.2  | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |
| ASO_4  | ASO2465_2_6 | ENST00000659788.1 | 102.2 | 102.2     | 102.2   | 102.2  | 102.2  | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |
| ASO_4  | ASO2465_2_6 | ENST00000710753.1 | 102.2 | 102.2     | 102.2   | 102.2  | 102.2  | Canonical RNA ViennaRNA model; MOE/PS/5m-dC not represented |

## 6 RBP CLIP, motif, expression, and function evidence

Sources referenced: ENCODE eCLIP, UCSC ENCODE tracks, POSTAR3, oRNAmnt, ATtRACT, RBPDB, RBP2GO (see provenance). **No CLIP peaks are invented.**

### 6.1 CLIP overlaps

| aso_id | clip_id     | clip_name   | clip_type   | clip_method | clip_version | clip_date   | clip_status | clip_notes  |
|--------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|
| ASO_4  | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6  | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6 |
| ASO_4  | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6  | ENC2465_2_6 | ENC2465_2_6 | ENC2465_2_6 |

asorlip\_sdpce\_data\_type\_peak\_peak\_kites\_isa\_vn\_kp\_0m14p20g1stsssbmgbfbrslnstataw\_a4areg

ASO NAE NC 2046 nonGRIN80Ds:API2026 metadGTP peak no peak overlap

|                                                                                                                                               |       |                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| //www.enco<br>deproj<br>ect.or<br>g/sear<br>ch/?ty<br>pe=E<br>xper<br>imen<br>t&ass<br>ay_tit<br>le=e<br>CLIP<br>&ass<br>embl<br>y=GR<br>Ch38 | 09-15 | BED files not<br>auto-<br>downloaded<br>for SHANK3<br>locus in this<br>run;<br>POSTAR3<br>requires<br>interac-<br>tive/query<br>UI; no peaks<br>invented ;<br>overlaps left<br>empty (not<br>fabricated) |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

ASO\_NAFC240006:GLP\_PESTANALYSIS:ACID\_ELISA non-GRCH38DE:API2026meta:GLP\_peakno\_peak\_overlap

```
//www.09-
enco15
deproj
ect.or
g/sear
ch/?ty
pe=E
xper
imen
t&ass
ay_tit
le=e
CLIP
&ass
embl
y=GR
Ch38
```

asorip\_clip\_sdrp\_celt\_asppick\_peak\_annotated\_overlap\_0911092010\_gintoss\_bed\_of\_the\_slow\_start\_date

ASO\_N4\_ENC2\_DE\_061115\_POSTAR3\_5565\_FNCSD5\_LNA nonGRCh38 DE:API\_2026 meta:ALT\_peak\_no\_peak\_overlap

//www. 09- BED files not  
enco 15 auto-  
deproj downloaded  
ect.or for SHANK3  
g/sear locus in this  
ch/?ty run;  
pe=E POSTAR3  
xper requires  
imen interac-  
t&ass tive/query  
ay\_tit UI; no peaks  
le=e invented ;  
CLIP overlaps left  
&ass empty (not  
embl fabricated)  
y=GR  
Ch38

ASO\_N4\_ENC2\_DE\_061115\_POSTAR3\_5565\_FNCSD5\_LNA nonGRCh38 DE:API\_2026 meta:ALT\_peak\_no\_peak\_overlap

//www. 09- BED files not  
enco 15 auto-  
deproj downloaded  
ect.or for SHANK3  
g/sear locus in this  
ch/?ty run;  
pe=E POSTAR3  
xper requires  
imen interac-  
t&ass tive/query  
ay\_tit UI; no peaks  
le=e invented ;  
CLIP overlaps left  
&ass empty (not  
embl fabricated)  
y=GR  
Ch38



| aso_clip_sdp_cel_tas_peak_peak_title_overlap_0m13p30gntossbedoftheslowstatadating |                                                                                                   |                                |                                                                                                                                                              |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASO_NA_ECDE_cel_tas_peak_peak_title_overlap_0m13p30gntossbedoftheslowstatadating  | nonRCN30DFs:API_2026                                                                              | metaDataP_peak_no_peak_overlap |                                                                                                                                                              |
|                                                                                   | //www. enco deproj ect.or g/sear ch/?ty pe=E xper imen t&ass ay_tit le=e CLIP &ass embl y=GR Ch38 | 09-15                          | BED files not auto-downloaded for SHANK3 locus in this run; POSTAR3 requires interac-tive/query UI; no peaks invented ; overlaps left empty (not fabricated) |

## 6.2 Motif overlaps (sample)

| aso_motif_sdp_cel_tas_peak_peak_title_overlap_0m13p30gntossbedoftheslowstatadating |             |                                                |  |
|------------------------------------------------------------------------------------|-------------|------------------------------------------------|--|
| ASO_F4_M12_4_Fp_M12_4_C71560_4_s5a2p4p1870000NA1478671ta539pA3E/2026               | style 09-15 | motif approx; not CLIP-validated at this locus |  |
| ASO_F4_M12_4_Fp_M12_4_C71560_4_s5a2p4p1870000NA1478671ta539pA3E/2026               | style 09-15 | motif approx; not CLIP-validated at this locus |  |
| ASO_F4_M12_4_Fp_M12_4_C71560_4_s5a2p4p1870000NA1478671ta539pA3E/2026               | style 09-15 | motif approx; not CLIP-validated at this locus |  |
| ASO_F4_M12_4_Fp_M12_4_C71560_4_s5a2p4p1870000NA1478671ta539pA3E/2026               | style 09-15 | motif approx; not CLIP-validated at this locus |  |
| ASO_F4_M12_4_Fp_M12_4_C71560_4_s5a2p4p1870000NA1478671ta539pA3E/2026               | style 09-15 | motif approx; not CLIP-validated at this locus |  |

| aso_id        | motif_source | motif_sequence | id    | date_created | date_modified | score | length | chr   | start | end  | strand | category          |
|---------------|--------------|----------------|-------|--------------|---------------|-------|--------|-------|-------|------|--------|-------------------|
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 52444 | 2013-05-01   | 2013-05-01    | 0.00  | 14     | 14786 | 7145  | 5308 | +      | Consensus         |
|               |              |                | 5816  |              |               |       |        |       |       |      |        | style motif       |
|               |              |                |       |              |               |       |        |       |       |      |        | con- 09- approx;  |
|               |              |                |       |              |               |       |        |       |       |      |        | sen- 15 not CLIP- |
|               |              |                |       |              |               |       |        |       |       |      |        | sus validated     |
|               |              |                |       |              |               |       |        |       |       |      |        | (cu- at this      |
|               |              |                |       |              |               |       |        |       |       |      |        | rated) locus      |
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 52444 | 2013-05-01   | 2013-05-01    | 0.00  | 14     | 14786 | 7145  | 5308 | +      | Consensus         |
|               |              |                | 5816  |              |               |       |        |       |       |      |        | style motif       |
|               |              |                |       |              |               |       |        |       |       |      |        | con- 09- approx;  |
|               |              |                |       |              |               |       |        |       |       |      |        | sen- 15 not CLIP- |
|               |              |                |       |              |               |       |        |       |       |      |        | sus validated     |
|               |              |                |       |              |               |       |        |       |       |      |        | (cu- at this      |
|               |              |                |       |              |               |       |        |       |       |      |        | rated) locus      |
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 52444 | 2013-05-01   | 2013-05-01    | 0.00  | 14     | 14786 | 7145  | 5308 | +      | Consensus         |
|               |              |                | 5816  |              |               |       |        |       |       |      |        | style motif       |
|               |              |                |       |              |               |       |        |       |       |      |        | con- 09- approx;  |
|               |              |                |       |              |               |       |        |       |       |      |        | sen- 15 not CLIP- |
|               |              |                |       |              |               |       |        |       |       |      |        | sus validated     |
|               |              |                |       |              |               |       |        |       |       |      |        | (cu- at this      |
|               |              |                |       |              |               |       |        |       |       |      |        | rated) locus      |
| ASO_FB_12_4   | RefSeq       | CTTGGT         | 52444 | 2013-05-01   | 2013-05-01    | 0.00  | 14     | 73971 | 2146  | 6078 | +      | Consensus         |
|               |              |                | 6512  |              |               |       |        |       |       |      |        | style motif       |
|               |              |                |       |              |               |       |        |       |       |      |        | con- 09- approx;  |
|               |              |                |       |              |               |       |        |       |       |      |        | sen- 15 not CLIP- |
|               |              |                |       |              |               |       |        |       |       |      |        | sus validated     |
|               |              |                |       |              |               |       |        |       |       |      |        | (cu- at this      |
|               |              |                |       |              |               |       |        |       |       |      |        | rated) locus      |
| ASO_FB_12_4   | RefSeq       | CTTGGT         | 52444 | 2013-05-01   | 2013-05-01    | 0.00  | 14     | 73971 | 2146  | 6078 | +      | Consensus         |
|               |              |                | 6512  |              |               |       |        |       |       |      |        | style motif       |
|               |              |                |       |              |               |       |        |       |       |      |        | con- 09- approx;  |
|               |              |                |       |              |               |       |        |       |       |      |        | sen- 15 not CLIP- |
|               |              |                |       |              |               |       |        |       |       |      |        | sus validated     |
|               |              |                |       |              |               |       |        |       |       |      |        | (cu- at this      |
|               |              |                |       |              |               |       |        |       |       |      |        | rated) locus      |
| ASO_FB_12_4   | RefSeq       | CTTGGT         | 52444 | 2013-05-01   | 2013-05-01    | 0.00  | 14     | 73971 | 2146  | 6078 | +      | Consensus         |
|               |              |                | 6512  |              |               |       |        |       |       |      |        | style motif       |
|               |              |                |       |              |               |       |        |       |       |      |        | con- 09- approx;  |
|               |              |                |       |              |               |       |        |       |       |      |        | sen- 15 not CLIP- |
|               |              |                |       |              |               |       |        |       |       |      |        | sus validated     |
|               |              |                |       |              |               |       |        |       |       |      |        | (cu- at this      |
|               |              |                |       |              |               |       |        |       |       |      |        | rated) locus      |
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 52444 | 2013-05-01   | 2013-05-01    | 0.00  | 14     | 14786 | 7145  | 5308 | +      | Consensus         |
|               |              |                | 6512  |              |               |       |        |       |       |      |        | style motif       |
|               |              |                |       |              |               |       |        |       |       |      |        | con- 09- approx;  |
|               |              |                |       |              |               |       |        |       |       |      |        | sen- 15 not CLIP- |
|               |              |                |       |              |               |       |        |       |       |      |        | sus validated     |
|               |              |                |       |              |               |       |        |       |       |      |        | (cu- at this      |
|               |              |                |       |              |               |       |        |       |       |      |        | rated) locus      |

| aso_id       | motif_source | motif_sequence | id      | date_created | date_modified | source  | length | how_created | status    |
|--------------|--------------|----------------|---------|--------------|---------------|---------|--------|-------------|-----------|
| ASO_RB_FOX14 | RefSeq       | GCAGAG         | 5app44  | EN1870000    | NA73071.2     | Ensembl | 600    | FASTA       | Consensus |
|              |              | 6512           |         |              |               |         |        | style       | 09-       |
|              |              |                |         |              |               |         |        | con-        | 15        |
|              |              |                |         |              |               |         |        | sen-        |           |
|              |              |                |         |              |               |         |        | sus         |           |
|              |              |                |         |              |               |         |        | (cu-        |           |
|              |              |                |         |              |               |         |        | rated)      |           |
|              |              |                |         |              |               |         |        |             | locus     |
| ASO_RB_FOX14 | RefSeq       | GCAGAG         | 5app44  | EN1870000    | NA73071.2     | Ensembl | 600    | FASTA       | Consensus |
|              |              | 6512           |         |              |               |         |        | style       | 09-       |
|              |              |                |         |              |               |         |        | con-        | 15        |
|              |              |                |         |              |               |         |        | sen-        |           |
|              |              |                |         |              |               |         |        | sus         |           |
|              |              |                |         |              |               |         |        | (cu-        |           |
|              |              |                |         |              |               |         |        | rated)      |           |
|              |              |                |         |              |               |         |        |             | locus     |
| ASO_RB_FOX14 | RefSeq       | GCAGAG         | 5app44  | EN1870000    | NA73071.2     | Ensembl | 600    | FASTA       | Consensus |
|              |              | 6512           |         |              |               |         |        | style       | 09-       |
|              |              |                |         |              |               |         |        | con-        | 15        |
|              |              |                |         |              |               |         |        | sen-        |           |
|              |              |                |         |              |               |         |        | sus         |           |
|              |              |                |         |              |               |         |        | (cu-        |           |
|              |              |                |         |              |               |         |        | rated)      |           |
|              |              |                |         |              |               |         |        |             | locus     |
| ASO_FB12_4   | MPIC         | CTTGGT         | 4sapp44 | EN1870000    | NA62795.7     | Ensembl | 525    | FASTA       | Consensus |
|              |              | 5682           |         |              |               |         |        | style       | 09-       |
|              |              |                |         |              |               |         |        | con-        | 15        |
|              |              |                |         |              |               |         |        | sen-        |           |
|              |              |                |         |              |               |         |        | sus         |           |
|              |              |                |         |              |               |         |        | (cu-        |           |
|              |              |                |         |              |               |         |        | rated)      |           |
|              |              |                |         |              |               |         |        |             | locus     |
| ASO_FB12_4   | MPIC         | CTTGGT         | 4sapp44 | EN1870000    | NA62795.7     | Ensembl | 525    | FASTA       | Consensus |
|              |              | 5682           |         |              |               |         |        | style       | 09-       |
|              |              |                |         |              |               |         |        | con-        | 15        |
|              |              |                |         |              |               |         |        | sen-        |           |
|              |              |                |         |              |               |         |        | sus         |           |
|              |              |                |         |              |               |         |        | (cu-        |           |
|              |              |                |         |              |               |         |        | rated)      |           |
|              |              |                |         |              |               |         |        |             | locus     |
| ASO_FB12_4   | MPIC         | CTTGGT         | 4sapp44 | EN1870000    | NA62795.7     | Ensembl | 525    | FASTA       | Consensus |
|              |              | 5682           |         |              |               |         |        | style       | 09-       |
|              |              |                |         |              |               |         |        | con-        | 15        |
|              |              |                |         |              |               |         |        | sen-        |           |
|              |              |                |         |              |               |         |        | sus         |           |
|              |              |                |         |              |               |         |        | (cu-        |           |
|              |              |                |         |              |               |         |        | rated)      |           |
|              |              |                |         |              |               |         |        |             | locus     |
| ASO_RB_FOX14 | RefSeq       | GCAGAG         | 5app44  | EN1870000    | NA62795.7     | Ensembl | 525    | FASTA       | Consensus |
|              |              | 5682           |         |              |               |         |        | style       | 09-       |
|              |              |                |         |              |               |         |        | con-        | 15        |
|              |              |                |         |              |               |         |        | sen-        |           |
|              |              |                |         |              |               |         |        | sus         |           |
|              |              |                |         |              |               |         |        | (cu-        |           |
|              |              |                |         |              |               |         |        | rated)      |           |
|              |              |                |         |              |               |         |        |             | locus     |

| aso_id        | motif_source | motif_sequence | id   | date_created        | date_modified       | score   | length | chr | start | end  | strand | category        |
|---------------|--------------|----------------|------|---------------------|---------------------|---------|--------|-----|-------|------|--------|-----------------|
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 5    | 2014-01-27 00:00:00 | 2014-01-27 00:00:00 | 2.62795 | 7      | 14  | 5357  | 5363 | +      | Consensus       |
|               |              |                | 5682 |                     |                     |         |        |     |       |      |        | style motif     |
|               |              |                |      |                     |                     |         |        |     |       |      |        | con- 15 approx; |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sen- not CLIP-  |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sus validated   |
|               |              |                |      |                     |                     |         |        |     |       |      |        | (cu- at this    |
|               |              |                |      |                     |                     |         |        |     |       |      |        | rated) locus    |
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 5    | 2014-01-27 00:00:00 | 2014-01-27 00:00:00 | 2.62795 | 7      | 14  | 5357  | 5363 | +      | Consensus       |
|               |              |                | 5682 |                     |                     |         |        |     |       |      |        | style motif     |
|               |              |                |      |                     |                     |         |        |     |       |      |        | con- 15 approx; |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sen- not CLIP-  |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sus validated   |
|               |              |                |      |                     |                     |         |        |     |       |      |        | (cu- at this    |
|               |              |                |      |                     |                     |         |        |     |       |      |        | rated) locus    |
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 5    | 2014-01-27 00:00:00 | 2014-01-27 00:00:00 | 2.62795 | 7      | 14  | 5357  | 5363 | +      | Consensus       |
|               |              |                | 5682 |                     |                     |         |        |     |       |      |        | style motif     |
|               |              |                |      |                     |                     |         |        |     |       |      |        | con- 15 approx; |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sen- not CLIP-  |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sus validated   |
|               |              |                |      |                     |                     |         |        |     |       |      |        | (cu- at this    |
|               |              |                |      |                     |                     |         |        |     |       |      |        | rated) locus    |
| ASO_FB_12_4   | RefSeq       | CTTGGG         | 4    | 2014-01-27 00:00:00 | 2014-01-27 00:00:00 | 4.45220 | 7      | 14  | 3357  | 3363 | +      | Consensus       |
|               |              |                | 4284 |                     |                     |         |        |     |       |      |        | style motif     |
|               |              |                |      |                     |                     |         |        |     |       |      |        | con- 15 approx; |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sen- not CLIP-  |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sus validated   |
|               |              |                |      |                     |                     |         |        |     |       |      |        | (cu- at this    |
|               |              |                |      |                     |                     |         |        |     |       |      |        | rated) locus    |
| ASO_FB_12_4   | RefSeq       | CTTGGG         | 4    | 2014-01-27 00:00:00 | 2014-01-27 00:00:00 | 4.45220 | 7      | 14  | 3357  | 3363 | +      | Consensus       |
|               |              |                | 4284 |                     |                     |         |        |     |       |      |        | style motif     |
|               |              |                |      |                     |                     |         |        |     |       |      |        | con- 15 approx; |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sen- not CLIP-  |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sus validated   |
|               |              |                |      |                     |                     |         |        |     |       |      |        | (cu- at this    |
|               |              |                |      |                     |                     |         |        |     |       |      |        | rated) locus    |
| ASO_FB_12_4   | RefSeq       | CTTGGG         | 4    | 2014-01-27 00:00:00 | 2014-01-27 00:00:00 | 4.45220 | 7      | 14  | 3357  | 3363 | +      | Consensus       |
|               |              |                | 4284 |                     |                     |         |        |     |       |      |        | style motif     |
|               |              |                |      |                     |                     |         |        |     |       |      |        | con- 15 approx; |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sen- not CLIP-  |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sus validated   |
|               |              |                |      |                     |                     |         |        |     |       |      |        | (cu- at this    |
|               |              |                |      |                     |                     |         |        |     |       |      |        | rated) locus    |
| ASO_RB_F07_14 | RefSeq       | GCAGAG         | 5    | 2014-01-27 00:00:00 | 2014-01-27 00:00:00 | 4.45220 | 7      | 14  | 3357  | 3363 | +      | Consensus       |
|               |              |                | 4284 |                     |                     |         |        |     |       |      |        | style motif     |
|               |              |                |      |                     |                     |         |        |     |       |      |        | con- 15 approx; |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sen- not CLIP-  |
|               |              |                |      |                     |                     |         |        |     |       |      |        | sus validated   |
|               |              |                |      |                     |                     |         |        |     |       |      |        | (cu- at this    |
|               |              |                |      |                     |                     |         |        |     |       |      |        | rated) locus    |

| aso_id       | motif | motif | motif | sequence | chr | start  | end    | strand | score | region | known   | status    | dating    |
|--------------|-------|-------|-------|----------|-----|--------|--------|--------|-------|--------|---------|-----------|-----------|
| ASO_RB_F0X34 | red   | RFHOC | ACGAG | 455app   | 12  | 170000 | 144522 | 0      | 7.1   | 3457RA | GE/RB20 | motif     | consensus |
|              |       |       |       | 4284     |     |        |        |        |       | style  | 09-15   | motif     |           |
|              |       |       |       |          |     |        |        |        |       | con-   |         | approx;   |           |
|              |       |       |       |          |     |        |        |        |       | sen-   |         | not CLIP- |           |
|              |       |       |       |          |     |        |        |        |       | sus    |         | validated |           |
|              |       |       |       |          |     |        |        |        |       | (cu-   |         | at this   |           |
|              |       |       |       |          |     |        |        |        |       | rated) |         | locus     |           |
| ASO_RB_F0X34 | red   | RFHOC | ACGAG | 455app   | 12  | 170000 | 144522 | 0      | 7.1   | 3457RA | GE/RB20 | motif     | consensus |
|              |       |       |       | 4284     |     |        |        |        |       | style  | 09-15   | motif     |           |
|              |       |       |       |          |     |        |        |        |       | con-   |         | approx;   |           |
|              |       |       |       |          |     |        |        |        |       | sen-   |         | not CLIP- |           |
|              |       |       |       |          |     |        |        |        |       | sus    |         | validated |           |
|              |       |       |       |          |     |        |        |        |       | (cu-   |         | at this   |           |
|              |       |       |       |          |     |        |        |        |       | rated) |         | locus     |           |
| ASO_RB_F0X34 | red   | RFHOC | ACGAG | 455app   | 12  | 170000 | 144522 | 0      | 7.1   | 3457RA | GE/RB20 | motif     | consensus |
|              |       |       |       | 4284     |     |        |        |        |       | style  | 09-15   | motif     |           |
|              |       |       |       |          |     |        |        |        |       | con-   |         | approx;   |           |
|              |       |       |       |          |     |        |        |        |       | sen-   |         | not CLIP- |           |
|              |       |       |       |          |     |        |        |        |       | sus    |         | validated |           |
|              |       |       |       |          |     |        |        |        |       | (cu-   |         | at this   |           |
|              |       |       |       |          |     |        |        |        |       | rated) |         | locus     |           |
| ASO_FB_F12   | red   | MRAC  | CTAGG | 455app   | 14  | 187000 | 106440 | 2      | 7.1   | 3457RA | GE/RB20 | motif     | consensus |
|              |       |       |       | 4178     |     |        |        |        |       | style  | 09-15   | motif     |           |
|              |       |       |       |          |     |        |        |        |       | con-   |         | approx;   |           |
|              |       |       |       |          |     |        |        |        |       | sen-   |         | not CLIP- |           |
|              |       |       |       |          |     |        |        |        |       | sus    |         | validated |           |
|              |       |       |       |          |     |        |        |        |       | (cu-   |         | at this   |           |
|              |       |       |       |          |     |        |        |        |       | rated) |         | locus     |           |
| ASO_FB_F12   | red   | MRAC  | CTAGG | 455app   | 14  | 187000 | 106440 | 2      | 7.1   | 3457RA | GE/RB20 | motif     | consensus |
|              |       |       |       | 4178     |     |        |        |        |       | style  | 09-15   | motif     |           |
|              |       |       |       |          |     |        |        |        |       | con-   |         | approx;   |           |
|              |       |       |       |          |     |        |        |        |       | sen-   |         | not CLIP- |           |
|              |       |       |       |          |     |        |        |        |       | sus    |         | validated |           |
|              |       |       |       |          |     |        |        |        |       | (cu-   |         | at this   |           |
|              |       |       |       |          |     |        |        |        |       | rated) |         | locus     |           |

### 6.3 Expression priors (neuron / NPC)

| aso_idbp  | expression_source | cell_type | expression_method          | expression_resource | resource_version | download_status   | date       | warning                                                                         |
|-----------|-------------------|-----------|----------------------------|---------------------|------------------|-------------------|------------|---------------------------------------------------------------------------------|
| ASO_#MRP2 | curated_neuron    | neuron    | TB_UFA_quantitative_RNAseq | Arista              | edited           | temperature_c0216 | prior_only | ModRBP<br>quantitative<br>ENCODE/GTEEx<br>measurement;<br>flag as prior<br>only |

| aso_id      | bp | expression_cell_type | expression_escort | expression_resource | version | download_status | darning                                                                            |
|-------------|----|----------------------|-------------------|---------------------|---------|-----------------|------------------------------------------------------------------------------------|
| ASO_ELAVL2  | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |
| ASO_ELAZL4  | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |
| ASO_BFOXL1  | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |
| ASO_BFOXL3  | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |
| ASO_PTBP2L4 | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |
| ASO_CELF1L4 | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |
| ASO_ATXN2L4 | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |
| ASO_TDP43L4 | 4  | rated_neurons        | neuron_TUFA       | quantitative_RNAseq | 09-15   | 2026-09-15      | ModBSP set<br>quantitative<br>ENCODE/GTEX<br>measurement;<br>flag as prior<br>only |

| aso_id       | bp           | expression_source | cell_type    | expression_source | expression_source | expression_source | expression_source | version | download_status | dating                                                         |
|--------------|--------------|-------------------|--------------|-------------------|-------------------|-------------------|-------------------|---------|-----------------|----------------------------------------------------------------|
| ASO_#NRPA2B1 | neuron       | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |
| ASO_#IS2A4   | rated_neuron | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |
| ASO_#UM24    | rated_neuron | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |
| ASO_#GVZ104  | rated_neuron | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |
| ASO_#PF124   | rated_neuron | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |
| ASO_#GO24    | rated_neuron | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |
| ASO_#G52B4   | rated_neuron | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |
| ASO_#SF24    | rated_neuron | TBUFA             | quantitative | RNAseq            | prior             | temperature_c     | 2026              | primary | ModBP set       | 09-15 quantitative ENCODE/GTEX measurement; flag as prior only |

| aso_id     | bp | expression_source | cell_type | expression_source | expression_source | units   | version | download | status        | warning |
|------------|----|-------------------|-----------|-------------------|-------------------|---------|---------|----------|---------------|---------|
| ASO_5P5F24 | 4  | curated_neuronal  | prior     | TRUBA             | quantitative      | RNA-seq | 2026    | curated  | Not RBP set   |         |
|            |    |                   |           |                   |                   |         | 09-15   |          | quantitative  |         |
|            |    |                   |           |                   |                   |         |         |          | ENCODE/GTEX   |         |
|            |    |                   |           |                   |                   |         |         |          | measurement;  |         |
|            |    |                   |           |                   |                   |         |         |          | flag as prior |         |
|            |    |                   |           |                   |                   |         |         |          | only          |         |
| ASO_4MR24  | 4  | curated_neuronal  | prior     | TRUBA             | quantitative      | RNA-seq | 2026    | curated  | Not RBP set   |         |
|            |    |                   |           |                   |                   |         | 09-15   |          | quantitative  |         |
|            |    |                   |           |                   |                   |         |         |          | ENCODE/GTEX   |         |
|            |    |                   |           |                   |                   |         |         |          | measurement;  |         |
|            |    |                   |           |                   |                   |         |         |          | flag as prior |         |
|            |    |                   |           |                   |                   |         |         |          | only          |         |
| ASO_4LAV24 | 4  | curated_neuronal  | prior     | TRUBA             | quantitative      | RNA-seq | 2026    | curated  | Not RBP set   |         |
|            |    |                   |           |                   |                   |         | 09-15   |          | quantitative  |         |
|            |    |                   |           |                   |                   |         |         |          | ENCODE/GTEX   |         |
|            |    |                   |           |                   |                   |         |         |          | measurement;  |         |
|            |    |                   |           |                   |                   |         |         |          | flag as prior |         |
|            |    |                   |           |                   |                   |         |         |          | only          |         |

## 6.4 Function annotation

| aso_id     | bp | function_source | cell_type | function_source | function_source | units | version | download | status      | warning |
|------------|----|-----------------|-----------|-----------------|-----------------|-------|---------|----------|-------------|---------|
| ASO_4MR24  | 4  | GO:BP           | curated   | neuronal        | candidate       | RBP2  | 2026    | curated  | Functional  |         |
|            |    | binding         |           | neuronal RBP;   |                 | s://  | 09-15   |          | labels are  |         |
|            |    | / trans-        |           | locus-specific  |                 | rbp2  |         |          | priors; not |         |
|            |    | lation          |           | CLIP not        |                 | go.d  |         |          | proof of    |         |
|            |    | regula-         |           | confirmed in    |                 | kfz.  |         |          | SHANK3 en-  |         |
|            |    | tion            |           | this run        |                 | de/   |         |          | gagement    |         |
| ASO_4LAV24 | 4  | GO:BP           | curated   | neuronal        | candidate       | RBP2  | 2026    | curated  | Functional  |         |
|            |    | binding         |           | neuronal RBP;   |                 | s://  | 09-15   |          | labels are  |         |
|            |    | / stabi-        |           | locus-specific  |                 | rbp2  |         |          | priors; not |         |
|            |    | lization        |           | CLIP not        |                 | go.d  |         |          | proof of    |         |
|            |    |                 |           | confirmed in    |                 | kfz.  |         |          | SHANK3 en-  |         |
|            |    |                 |           | this run        |                 | de/   |         |          | gagement    |         |
| ASO_4LAV24 | 4  | GO:BP           | curated   | neuronal        | candidate       | RBP2  | 2026    | curated  | Functional  |         |
|            |    | binding         |           | neuronal RBP;   |                 | s://  | 09-15   |          | labels are  |         |
|            |    | / stabi-        |           | locus-specific  |                 | rbp2  |         |          | priors; not |         |
|            |    | lization        |           | CLIP not        |                 | go.d  |         |          | proof of    |         |
|            |    |                 |           | confirmed in    |                 | kfz.  |         |          | SHANK3 en-  |         |
|            |    |                 |           | this run        |                 | de/   |         |          | gagement    |         |
| ASO_4B5024 | 4  | GO:BP           | curated   | neuronal        | candidate       | RBP2  | 2026    | curated  | Functional  |         |
|            |    | regula-         |           | neuronal RBP;   |                 | s://  | 09-15   |          | labels are  |         |
|            |    | tion            |           | locus-specific  |                 | rbp2  |         |          | priors; not |         |
|            |    |                 |           | CLIP not        |                 | go.d  |         |          | proof of    |         |
|            |    |                 |           | confirmed in    |                 | kfz.  |         |          | SHANK3 en-  |         |
|            |    |                 |           | this run        |                 | de/   |         |          | gagement    |         |



| aso_id      | bp | function | molecular_function    | cellular_function | regulation        | development       | shank2    | allele                                                      | source | source_url | download   | status  | date       | comment                                           |
|-------------|----|----------|-----------------------|-------------------|-------------------|-------------------|-----------|-------------------------------------------------------------|--------|------------|------------|---------|------------|---------------------------------------------------|
| ASO_4B5_2B2 | 60 | Splicing | regulation (neuronal) | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |
| ASO_4T1_2B2 | 60 | Splicing | repression (neuronal) | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |
| ASO_4E5_2B2 | 60 | Splicing | / translation         | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |
| ASO_4T8_2B2 | 60 | Splicing | granule / translation | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |
| ASO_4D5_2B2 | 60 | Splicing | / RNA stability       | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |
| ASO_4N5_2B2 | 60 | Splicing | / processing          | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |
| ASO_4N5_2B2 | 60 | Splicing | / translation         | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |
| ASO_4U5_2B2 | 60 | Splicing | repression            | splicing          | neurotransmission | neurotransmission | Candidate | neuronal RBP; locus-specific CLIP not confirmed in this run | RBP2   | 60         | 2026-09-15 | curated | Functional | labels are priors; not proof of SHANK3 engagement |

| aso_id       | function                                  | method  | source        | status    | date                         | comment    |                                                                      |
|--------------|-------------------------------------------|---------|---------------|-----------|------------------------------|------------|----------------------------------------------------------------------|
| ASO_405_RBP2 | 5e style / RNAi-related                   | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |
| ASO_405_RBP2 | 5e style surveillance                     | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |
| ASO_405_RBP2 | 5e style miRNA repression                 | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |
| ASO_405_RBP2 | 5e style reader / stabilization           | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |
| ASO_405_RBP2 | 5e style pro-processing                   | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |
| ASO_405_RBP2 | 5e style activation                       | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |
| ASO_405_RBP2 | 5e style binding / translation regulation | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |
| ASO_405_RBP2 | 5e style binding / stabilization          | curated | neural injury | Candidate | RBP2 GO s://rbp2.go.dkfz.de/ | 2026-09-15 | curated Functional labels are priors; not proof of SHANK3 engagement |

## 7 Ranked RBP hypotheses

Transparent scoring: **+3 / +2 / +1 / -2** rules in R/06\_mechanism\_ranking.R. Top 3-5 RBPs per ASO attempt to include repressive, activating/stabilizing, and structure-related classes.

| aso_id       | rank | rbp_score | rbp_class  | regulatory_classification                                                                                                                                                                        | warning                                                        |
|--------------|------|-----------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| ASO_1_5P2B42 | 1    | 2         | repressive | repressive expression prior   +2 repressive function (occlusion could increase expression)   +1 splicing/processing                                                                              | Scores are transparent heuristics; not experimental validation |
| ASO_1_5P2B42 | 2    | 1         | repressive | repressive expression prior   +2 repressive function (occlusion could increase expression)   +1 splicing/processing                                                                              | Scores are transparent heuristics; not experimental validation |
| ASO_2_5P2B40 | 1    | 2         | X1splicing | context-dependent expression   +1 splicing/processing                                                                                                                                            | Scores are transparent heuristics; not experimental validation |
| ASO_2_5P2B40 | 2    | 1         | X1splicing | context-dependent expression   +1 splicing/processing                                                                                                                                            | Scores are transparent heuristics; not experimental validation |
| ASO_3_5P2B40 | 1    | 2         | X3splicing | context-dependent expression   +1 splicing/processing                                                                                                                                            | Scores are transparent heuristics; not experimental validation |
| ASO_3_5P2B40 | 2    | 1         | X3splicing | context-dependent expression   +1 splicing/processing                                                                                                                                            | Scores are transparent heuristics; not experimental validation |
| ASO_4_5P2B41 | 1    | 2         | activating | stabilizing expression prior   +2 activating/stabilizing function (phenotype-compatible if occlusion relieves repression OR displacement of repressor — context needed)   +1 splicing/processing | Scores are transparent heuristics; not experimental validation |
| ASO_4_5P2B41 | 2    | 1         | activating | stabilizing expression prior   +2 activating/stabilizing function (phenotype-compatible if occlusion relieves repression OR displacement of repressor — context needed)   +1 splicing/processing | Scores are transparent heuristics; not experimental validation |
| ASO_5_5A2C42 | 1    | 2         | repressive | repressive expression prior   +2 repressive function (occlusion could increase expression)                                                                                                       | Scores are transparent heuristics; not experimental validation |

| aso_id       | rank | bp | score | rbp_class  | regulatory | direction | hint                                                                            | warning                                                        |
|--------------|------|----|-------|------------|------------|-----------|---------------------------------------------------------------------------------|----------------------------------------------------------------|
| ASO_3_5A2_02 | 62   |    |       | repressive | repressive | 1         | expression prior   +2 repressive function (occlusion could increase expression) | Scores are transparent heuristics; not experimental validation |

## 8 TF and chromatin (conditional)

TF motifs are evaluated on **genomic DNA only** when locus class suggests nuclear regulatory / intronic / promoter / enhancer / antisense context. This evidence stream is separate from RBP analysis.

| aso_id        | tf | motif | chr | start | end | strand | score | rbp | class | evidence_note                                                                                                                          | genome | res  | build | date    | warning                                                            |
|---------------|----|-------|-----|-------|-----|--------|-------|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------|--------|------|-------|---------|--------------------------------------------------------------------|
| ASO_1A_1A2_1A | NA | NA    | NA  | NA    | NA  | NA     | NA    | NA  | NA    | TF/chromatin analysis skipped: target context not classified as nuclear regulatory/enhancer/promoter/antisense/intronic regulatory RNA | GRCm38 | 2026 | 09-15 | skipped | NoTFRegulatory_context inference on RNA; step gated by locus class |
| ASO_1A_1A2_1A | NA | NA    | NA  | NA    | NA  | NA     | NA    | NA  | NA    | TF/chromatin analysis skipped: target context not classified as nuclear regulatory/enhancer/promoter/antisense/intronic regulatory RNA | GRCm38 | 2026 | 09-15 | skipped | NoTFRegulatory_context inference on RNA; step gated by locus class |

## 9 Chemistry-aware mechanism ranking

Eight chemistry/mechanism rules are applied (architecture unknown; RNase-H unknown; phenotype direction; PS protein engagement; MOE occupancy bias; 5m-dC caveats; structure hypothetical; sequence≠proof). Labels scored as hypotheses only.

| aso_id        | rank | mechanism | hypothesis | base    | context                                | rank | ratio | caveat                                                |
|---------------|------|-----------|------------|---------|----------------------------------------|------|-------|-------------------------------------------------------|
| ASO_1A_1A2_1A | 4    | occlusion | 8          | unknown | WHE PTBP2, RBFOX1, RBFOX3, SRSF1, AGO2 | 1    | 0.001 | Hypothesis only. Do not treat as confirmed mechanism. |

| aso_rank       | mechanism     | hypothesis_rank | hypothesis_status | hypothesis_label | hypothesis_text                                                                                                               | caveat                                                |
|----------------|---------------|-----------------|-------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| ASO_2_RNA4     | stabilization | 3               | unknown           | TRUE             | PTBP2, RBFOX1, RBFOX3, SRP51, AGO2 (stabilization)    +1 Occupancy mechanisms can stabilize RNA in some contexts (hypothesis) | Hypothesis only. Do not treat as confirmed mechanism. |
| ASO_3_Splicing | or processing | 3               | unknown           | TRUE             | PTBP2, RBFOX1, RBFOX3, SRP51, AGO2                                                                                            | Hypothesis only. Do not treat as confirmed mechanism. |
| ASO_4_Splicing | resolved      | 3               | unknown           | TRUE             | PTBP2, RBFOX1, RBFOX3, SRP51, AGO2                                                                                            | Hypothesis only. Do not treat as confirmed mechanism. |
| ASO_5_RBP4     | displacement  | 2               | unknown           | TRUE             | PTBP2, RBFOX1, RBFOX3, SRP51, AGO2                                                                                            | Hypothesis only. Do not treat as confirmed mechanism. |
| ASO_14_RBP6    | occlusion     | 8               | unknown           | TRUE             | PTBP2, RBFOX1, RBFOX3, SRP51, AGO2                                                                                            | Hypothesis only. Do not treat as confirmed mechanism. |
| ASO_2_RNA6     | stabilization | 3               | unknown           | TRUE             | PTBP2, RBFOX1, RBFOX3, SRP51, AGO2 (stabilization)    +1 Occupancy mechanisms can stabilize RNA in some contexts (hypothesis) | Hypothesis only. Do not treat as confirmed mechanism. |
| ASO_3_Splicing | or processing | 3               | unknown           | TRUE             | PTBP2, RBFOX1, RBFOX3, SRP51, AGO2                                                                                            | Hypothesis only. Do not treat as confirmed mechanism. |

| aso_id      | rank | mechanism    | hypothesis | not_tested | shank3_match                       | rbp_rationale                                                                                     | caveat                                                |
|-------------|------|--------------|------------|------------|------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| ASO_4_5_2_6 | 3    | solved       | unknown    | TRUE       | PTBP2, RBFOX1, RBFOX3, SRSF1, AGO2 | Recognition vs unmodified models    +2 Rule8: sequence/DB evidence alone cannot confirm mechanism | Hypothesis only. Do not treat as confirmed mechanism. |
| ASO_4_5_2_6 | 2    | displacement | unknown    | TRUE       | PTBP2, RBFOX1, RBFOX3, SRSF1, AGO2 | Displacement RBP complexes                                                                        | Hypothesis only. Do not treat as confirmed mechanism. |

## 10 Per-ASO interpretation

### 10.1 ASO\_4\_5\_2\_4 (TCCTGACTGTGCGCTGGG)

- Top mechanism hypothesis: **RBP occlusion**
- Top RBPs: PTBP2, RBFOX1, RBFOX3, SRSF1, AGO2
- RNase-H: **unknown** (unsupported by chemistry assumption; architecture unknown; phenotype argues against simple on-target RNase-H knockdown)
- Exact SHANK3 match present: TRUE

### 10.2 ASO\_4\_5\_2\_6 (AGGGCACTAAGCAATGTA)

- Top mechanism hypothesis: **RBP occlusion**
- Top RBPs: PTBP2, RBFOX1, RBFOX3, SRSF1, AGO2
- RNase-H: **unknown** (unsupported by chemistry assumption; architecture unknown; phenotype argues against simple on-target RNase-H knockdown)
- Exact SHANK3 match present: TRUE

## 11 Off-target considerations

A limited **chr22 genomic exact reverse-complement** screen is recorded in `results/02_offtargets_chr22.ts`. This is **not** a full transcriptome off-target proof. Off-target mechanisms remain possible, especially without exact SHANK3 transcript confirmation.

## 12 Validation recommendations

For every high-priority RBP, the following ten experiment types are recommended (each states which hypothesis it tests):

| aso_id | rbp_rank | hypothesis_context                                          | experiment                                                                         | hypothesis_tested                                                                                            |
|--------|----------|-------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | ASO + RBP CLIP / eCLIP (or RIP-qPCR) at the mapped site                            | Tests whether RBP occupies the ASO site and whether ASO alters occupancy (occlusion/displacement)            |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RBP knockdown / CRISPRi and measure SHANK3 RNA/protein (directionality)            | Tests whether RBP is necessary for basal SHANK3 repression/activation matching ASO phenotype                 |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RBP overexpression rescue of ASO phenotype                                         | Tests sufficiency / epistasis of RBP relative to ASO                                                         |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RNA immunoprecipitation after ASO to test occlusion vs recruitment                 | Distinguishes occlusion vs recruitment of the RBP                                                            |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Competitive EMSA / binding assays with PS/MOE/5m-dC ASO chemistry                  | Tests direct ASO-RBP or ASO-RNA-RBP competition with matched chemistry                                       |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | DMS-MaPseq or SHAPE-MaP ± ASO for structure remodeling                             | Tests RNA-structure remodeling hypothesis at the site                                                        |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RNase-H cleavage assay in vitro with cognate chemistry (architecture test)         | Tests whether chemistry/architecture supports RNase-H (expected unknown/unsupported unless gapmer confirmed) |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Nascent transcription (GRO-seq / TT-seq) if transcriptional mechanism hypothesized | Tests transcriptional/enhancer regulation hypothesis                                                         |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Isoform-resolved RNA-seq / splicing reporter if splicing RBP hypothesized          | Tests splicing or processing hypothesis                                                                      |
| ASO_4  | PTBP2    | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Polysome profiling / ribosome footprinting if translation regulation hypothesized  | Tests translation regulation hypothesis                                                                      |

| aso_id | rbp_rank | hypothesis | context                                                     | experiment                                                                         | hypothesis_tested                                                                                            |
|--------|----------|------------|-------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | ASO + RBP CLIP / eCLIP (or RIP-qPCR) at the mapped site                            | Tests whether RBP occupies the ASO site and whether ASO alters occupancy (occlusion/displacement)            |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RBP knockdown / CRISPRi and measure SHANK3 RNA/protein (directionality)            | Tests whether RBP is necessary for basal SHANK3 repression/activation matching ASO phenotype                 |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RBP overexpression rescue of ASO phenotype                                         | Tests sufficiency / epistasis of RBP relative to ASO                                                         |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RNA immunoprecipitation after ASO to test occlusion vs recruitment                 | Distinguishes occlusion vs recruitment of the RBP                                                            |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Competitive EMSA / binding assays with PS/MOE/5m-dC ASO chemistry                  | Tests direct ASO-RBP or ASO-RNA-RBP competition with matched chemistry                                       |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | DMS-MaPseq or SHAPE-MaP ± ASO for structure remodeling                             | Tests RNA-structure remodeling hypothesis at the site                                                        |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | RNase-H cleavage assay in vitro with cognate chemistry (architecture test)         | Tests whether chemistry/architecture supports RNase-H (expected unknown/unsupported unless gapmer confirmed) |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Nascent transcription (GRO-seq / TT-seq) if transcriptional mechanism hypothesized | Tests transcriptional/enhancer regulation hypothesis                                                         |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Isoform-resolved RNA-seq / splicing reporter if splicing RBP hypothesized          | Tests splicing or processing hypothesis                                                                      |
| ASO_4  | PTBP2    | 264        | ASO engages PTBP2 pathway (repressive; repressive_splicing) | Polysome profiling / ribosome footprinting if translation regulation hypothesized  | Tests translation regulation hypothesis                                                                      |



| aso_id      | rbp_rank | hypothesis_context                                       | experiment                                                                         | hypothesis_tested                                                                                            |
|-------------|----------|----------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| ASO_4BF0X14 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | ASO + RBP CLIP / eCLIP (or RIP-qPCR) at the mapped site                            | Tests whether RBP occupies the ASO site and whether ASO alters occupancy (occlusion/displacement)            |
| ASO_4BF0X14 | 2        | ASO engages RBFOX1 pathway (splicing; context_dependent) | RBP knockdown / CRISPRi and measure SHANK3 RNA/protein (directionality)            | Tests whether RBP is necessary for basal SHANK3 repression/activation matching ASO phenotype                 |
| ASO_4BF0X14 | 3        | ASO engages RBFOX1 pathway (splicing; context_dependent) | RBP overexpression rescue of ASO phenotype                                         | Tests sufficiency / epistasis of RBP relative to ASO                                                         |
| ASO_4BF0X14 | 4        | ASO engages RBFOX1 pathway (splicing; context_dependent) | RNA immunoprecipitation after ASO to test occlusion vs recruitment                 | Distinguishes occlusion vs recruitment of the RBP                                                            |
| ASO_4BF0X14 | 5        | ASO engages RBFOX1 pathway (splicing; context_dependent) | Competitive EMSA / binding assays with PS/MOE/5m-dC ASO chemistry                  | Tests direct ASO-RBP or ASO-RNA-RBP competition with matched chemistry                                       |
| ASO_4BF0X14 | 6        | ASO engages RBFOX1 pathway (splicing; context_dependent) | DMS-MaPseq or SHAPE-MaP ± ASO for structure remodeling                             | Tests RNA-structure remodeling hypothesis at the site                                                        |
| ASO_4BF0X14 | 7        | ASO engages RBFOX1 pathway (splicing; context_dependent) | RNase-H cleavage assay in vitro with cognate chemistry (architecture test)         | Tests whether chemistry/architecture supports RNase-H (expected unknown/unsupported unless gapmer confirmed) |
| ASO_4BF0X14 | 8        | ASO engages RBFOX1 pathway (splicing; context_dependent) | Nascent transcription (GRO-seq / TT-seq) if transcriptional mechanism hypothesized | Tests transcriptional/enhancer regulation hypothesis                                                         |
| ASO_4BF0X14 | 9        | ASO engages RBFOX1 pathway (splicing; context_dependent) | Isoform-resolved RNA-seq / splicing reporter if splicing RBP hypothesized          | Tests splicing or processing hypothesis                                                                      |
| ASO_4BF0X14 | 10       | ASO engages RBFOX1 pathway (splicing; context_dependent) | Polysome profiling / ribosome footprinting if translation regulation hypothesized  | Tests translation regulation hypothesis                                                                      |
| ASO_4BF0X64 | 11       | ASO engages RBFOX1 pathway (splicing; context_dependent) | ASO + RBP CLIP / eCLIP (or RIP-qPCR) at the mapped site                            | Tests whether RBP occupies the ASO site and whether ASO alters occupancy (occlusion/displacement)            |
| ASO_4BF0X64 | 12       | ASO engages RBFOX1 pathway (splicing; context_dependent) | RBP knockdown / CRISPRi and measure SHANK3 RNA/protein (directionality)            | Tests whether RBP is necessary for basal SHANK3 repression/activation matching ASO phenotype                 |

| aso_id      | rbp_rank | hypothesis_context                                       | experiment                                                                         | hypothesis_tested                                                                                            |
|-------------|----------|----------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | RBP overexpression rescue of ASO phenotype                                         | Tests sufficiency / epistasis of RBP relative to ASO                                                         |
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | RNA immunoprecipitation after ASO to test occlusion vs recruitment                 | Distinguishes occlusion vs recruitment of the RBP                                                            |
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | Competitive EMSA / binding assays with PS/MOE/5m-dC ASO chemistry                  | Tests direct ASO-RBP or ASO-RNA-RBP competition with matched chemistry                                       |
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | DMS-MaPseq or SHAPE-MaP ± ASO for structure remodeling                             | Tests RNA-structure remodeling hypothesis at the site                                                        |
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | RNase-H cleavage assay in vitro with cognate chemistry (architecture test)         | Tests whether chemistry/architecture supports RNase-H (expected unknown/unsupported unless gapmer confirmed) |
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | Nascent transcription (GRO-seq / TT-seq) if transcriptional mechanism hypothesized | Tests transcriptional/enhancer regulation hypothesis                                                         |
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | Isoform-resolved RNA-seq / splicing reporter if splicing RBP hypothesized          | Tests splicing or processing hypothesis                                                                      |
| ASO_4BF0X64 | 1        | ASO engages RBFOX1 pathway (splicing; context_dependent) | Polysome profiling / ribosome footprinting if translation regulation hypothesized  | Tests translation regulation hypothesis                                                                      |
| ASO_4BF0X64 | 1        | ASO engages RBFOX3 pathway (splicing; context_dependent) | ASO + RBP CLIP / eCLIP (or RIP-qPCR) at the mapped site                            | Tests whether RBP occupies the ASO site and whether ASO alters occupancy (occlusion/displacement)            |
| ASO_4BF0X64 | 1        | ASO engages RBFOX3 pathway (splicing; context_dependent) | RBP knockdown / CRISPRi and measure SHANK3 RNA/protein (directionality)            | Tests whether RBP is necessary for basal SHANK3 repression/activation matching ASO phenotype                 |
| ASO_4BF0X64 | 1        | ASO engages RBFOX3 pathway (splicing; context_dependent) | RBP overexpression rescue of ASO phenotype                                         | Tests sufficiency / epistasis of RBP relative to ASO                                                         |
| ASO_4BF0X64 | 1        | ASO engages RBFOX3 pathway (splicing; context_dependent) | RNA immunoprecipitation after ASO to test occlusion vs recruitment                 | Distinguishes occlusion vs recruitment of the RBP                                                            |

| aso_id      | rbp                                                      | hypothesis                                                                         | experiment                                                                                                   | hypothesis_tested |
|-------------|----------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------|
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Competitive EMSA / binding assays with PS/MOE/5m-dC ASO chemistry                  | Tests direct ASO-RBP or ASO-RNA-RBP competition with matched chemistry                                       |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | DMS-MaPseq or SHAPE-MaP ± ASO for structure remodeling                             | Tests RNA-structure remodeling hypothesis at the site                                                        |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | RNase-H cleavage assay in vitro with cognate chemistry (architecture test)         | Tests whether chemistry/architecture supports RNase-H (expected unknown/unsupported unless gapmer confirmed) |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Nascent transcription (GRO-seq / TT-seq) if transcriptional mechanism hypothesized | Tests transcriptional/enhancer regulation hypothesis                                                         |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Isoform-resolved RNA-seq / splicing reporter if splicing RBP hypothesized          | Tests splicing or processing hypothesis                                                                      |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Polysome profiling / ribosome footprinting if translation regulation hypothesized  | Tests translation regulation hypothesis                                                                      |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | ASO + RBP CLIP / eCLIP (or RIP-qPCR) at the mapped site                            | Tests whether RBP occupies the ASO site and whether ASO alters occupancy (occlusion/displacement)            |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | RBP knockdown / CRISPRi and measure SHANK3 RNA/protein (directionality)            | Tests whether RBP is necessary for basal SHANK3 repression/activation matching ASO phenotype                 |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | RBP overexpression rescue of ASO phenotype                                         | Tests sufficiency / epistasis of RBP relative to ASO                                                         |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | RNA immunoprecipitation after ASO to test occlusion vs recruitment                 | Distinguishes occlusion vs recruitment of the RBP                                                            |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Competitive EMSA / binding assays with PS/MOE/5m-dC ASO chemistry                  | Tests direct ASO-RBP or ASO-RNA-RBP competition with matched chemistry                                       |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | DMS-MaPseq or SHAPE-MaP ± ASO for structure remodeling                             | Tests RNA-structure remodeling hypothesis at the site                                                        |                   |

| aso_id      | rbp_rank                                                 | hypothesis_context                                                                 | experiment                                                                                                   | hypothesis_tested |
|-------------|----------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------|
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | RNase-H cleavage assay in vitro with cognate chemistry (architecture test)         | Tests whether chemistry/architecture supports RNase-H (expected unknown/unsupported unless gapmer confirmed) |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Nascent transcription (GRO-seq / TT-seq) if transcriptional mechanism hypothesized | Tests transcriptional/enhancer regulation hypothesis                                                         |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Isoform-resolved RNA-seq / splicing reporter if splicing RBP hypothesized          | Tests splicing or processing hypothesis                                                                      |                   |
| ASO_4BF0X64 | ASO engages RBFOX3 pathway (splicing; context_dependent) | Polysome profiling / ribosome footprinting if translation regulation hypothesized  | Tests translation regulation hypothesis                                                                      |                   |

## 13 Resource provenance

| resource          | version             | url                                                                                                                                                                                                                                                                     | genome_build | chr | chr_start | chr_end    | date     | time_utc |
|-------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----|-----------|------------|----------|----------|
| GENCODE_GTF       | GENCODE Re-lease 45 | <a href="https://ftp.ebi.ac.uk/pub/data/bases/gencode/Gencode_human/release_45/annotation.gtf.gz">https://ftp.ebi.ac.uk/pub/data/bases/gencode/Gencode_human/release_45/annotation.gtf.gz</a>                                                                           | GRCh38       | 26  | 09-15     | 2026-09-15 | 14:00:41 | UTC      |
| TxDb.Hsapiens     | 1.11.0              | BSgenome.Hsapiens                                                                                                                                                                                                                                                       | GRCh38       | 26  | 09-15     | 2026-09-15 | 14:00:41 | UTC      |
| BSgenome.Hsapiens | 1.11.0              | BSgenome.Hsapiens                                                                                                                                                                                                                                                       | GRCh38       | 26  | 09-15     | 2026-09-15 | 14:00:41 | UTC      |
| ENCODE_eCLIP      | ENCODE Portal API   | <a href="https://www.encodeproject.org/search/?type=Experiment&amp;assay_title=eCLIP&amp;assembly=GRCh38&amp;format=json&amp;limit=50">https://www.encodeproject.org/search/?type=Experiment&amp;assay_title=eCLIP&amp;assembly=GRCh38&amp;format=json&amp;limit=50</a> | GRCh38       | 26  | 09-15     | 2026-09-15 | 14:00:58 | UTC      |
| POSTAR3           | POSTAR3             | <a href="http://postar.ncrnlab.org/">http://postar.ncrnlab.org/</a>                                                                                                                                                                                                     | GRCh38       | 26  | 09-15     | 2026-09-15 | 14:00:58 | UTC      |
| oRNAMENT          | oRNAMENT            | <a href="https://rnabio.ircm.qc.ca/oRNAMENT/">https://rnabio.ircm.qc.ca/oRNAMENT/</a>                                                                                                                                                                                   | GRCh38       | 26  | 09-15     | 2026-09-15 | 14:00:58 | UTC      |

| resource     | version               | url                                                                         | genome_build | build_date | data_date  | time_utc                                 |
|--------------|-----------------------|-----------------------------------------------------------------------------|--------------|------------|------------|------------------------------------------|
| ATtRACT      | ATtRACT               | <a href="https://attract.cnice.es/">https://attract.cnice.es/</a>           | GRCh38       | 2026-09-15 | 2026-09-15 | Motif resource referenced                |
| RBPDB        | RBPDB                 | <a href="http://rbpdb.ccbr.utoronto.ca/">http://rbpdb.ccbr.utoronto.ca/</a> | GRCh38       | 2026-09-15 | 2026-09-15 | Motif resource referenced                |
| RBP2GO       | RBP2GO                | <a href="https://rbp2go.dkfz.de/">https://rbp2go.dkfz.de/</a>               | GRCh38       | 2026-09-15 | 2026-09-15 | Function annotation resource referenced  |
| ENCODE_TF    | ENCODE                | <a href="https://www.encodeproject.org/">https://www.encodeproject.org/</a> | GRCh38       | 2026-09-15 | 2026-09-15 | TF/chromatin step conditional            |
| ChIP-Atlas   | ChIP-Atlas            | <a href="https://chip-atlas.org/">https://chip-atlas.org/</a>               | GRCh38       | 2026-09-15 | 2026-09-15 | Referenced; peaks not fabricated         |
| SCREEN_cCREs | SCREEN / ENCODE cCREs | <a href="https://screen.wenglab.org/">https://screen.wenglab.org/</a>       | GRCh38       | 2026-09-15 | 2026-09-15 | Referenced for enhancer/promoter context |

## 14 Limitations and missing information

- ASO **architecture** not provided (cannot confirm gapmer / mixmer / fully modified).
- **RNase-H compatibility unknown**; not assumed.
- Neuron/NPC **eCLIP peak overlaps** at the precise site generally unavailable in this automated run (metadata may be cached; peaks not fabricated).
- Motif/expression layers include **explicitly labeled priors**.
- Structure values omit MOE/PS/5m-dC parameterization.
- No experimental ASO engagement assay integrated.
- Genome builds other than GRCh38 were **not** used.

## 15 Conclusions (non-confirmatory)

Both ASOs should be interpreted under chemistry-aware occupancy / RBP / structure / splicing hypotheses rather than assumed RNase-H gapmer knockdown. The highest-scoring mechanism labels are **hypotheses only**. Definitive mechanism assignment requires the validation suite above, architecture disclosure, and locus-resolved CLIP/structure/functional genetics in neuron/NPC systems.

## 16 Reproducibility

```
cd shank3-aso-mechanism
Rscript -e 'targets::tar_make()'
```

Pipeline orchestrated with the `targets` package; modules in `R/01_*.R-R/07_*.R`; config in `_config.yml`.